

Dokumentowe bazy danych – MongoDB

Ćwiczenie 1

Imiona i nazwiska autorów:

Test połączenia

- informacja o wersji

```
db.version()
```

- wynik

```
{
  "result": "7.0.7"
}
```

-
- lista baz danych na serwerze

```
show dbs;
show databases;
db.adminCommand('listDatabases');
```

- wynik

```
[
  {
    "empty": false,
    "name": "admin",
    "sizeOnDisk": 40960
  },
  {
    "empty": false,
    "name": "config",
    "sizeOnDisk": 98304
  },
  {
    "empty": false,
    "name": "local",

```

```
    "sizeOnDisk": 40960
  }
]
```

- wybór bazy danych
 - baza o nazwie univ

```
use univ;
```

- informacja o bieżącej bazie danych

```
db;
```

- informacja o kolekcjach

```
show collections;
db.getCollectionNames();
```

Proste operacje/zapytania

- MongoDB simple query
- <https://www.mongodb.com/docs/manual/tutorial/query-documents/>

Przykład 1

proste operacje na dokumentach

- wstaw/stwórz pierwszy dokument
 - w tym momencie tworzona jest baza danych i kolekcja (jeśli wcześniej nie istniały)

```
db.student.insertOne(
  {
    "student_id": 1,
    "firstname": "John",
    "lastname": "Gold",
    "age": 25
  }
);
```

- sprawdź listę baz i kolekcji

```
show databases;
```

- wynik

```
[
  {
    "empty": false,
    "name": "admin",
    "sizeOnDisk": 40960
  },
  {
    "empty": false,
    "name": "config",
    "sizeOnDisk": 98304
  },
  {
    "empty": false,
    "name": "local",
    "sizeOnDisk": 40960
  },
  {
    "empty": false,
    "name": "univ",
    "sizeOnDisk": 8192
  }
]
```

```
show collections;
```

- wynik

```
[
  {
    "badge": "",
    "name": "student"
  }
]
```

- wyszukaj dokumenty w kolekcji **employees**

```
db.student.find();
db.student.find({});
```

- wynik

```
[
  {
    "_id": {"$oid": "67e86e45b1c69e3d7e42c9c0"},
    "age": 25,
    "firstname": "John",
    "lastname": "Gold",
    "student_id": 1
  }
]
```

- stwórz indeks zapewniający unikalność atrybutu "student_id"

```
// tworzenie indeksu
db.student.createIndex({ "student_id" : 1 }, { "unique": true });

// inf. o indeksach w kolekcji student
db.student.getIndexes();

// usunięcie indeksu o nazwie
db.student.dropIndexes("student_id_1");
```

- wstaw kilka dokumentów do kolekcji student
 - np

```
db.student.insertOne(
  {
    "student_id": 2,
    "firstname": "James",
    "lastname": "Bond"
  }
);

db.student.insertOne(
  {
    "firstname": "John",
    "lastname": "Bond"
  }
);
```

- wyszukaj dokumenty w kolekcji student

```
db.student.find();
db.student.find({});
db.student.find({"firstname": "John"});
db.student.find({"student_id": 1});

db.student.find({"_id": ObjectId("67e669647013d10c86e71c87")});
```

- zmodyfikuj wybrane dokumenty
 - po wykonaniu każdego z przykładów sprawdź wynik za pomocą `db.student.find()`
- np

```
db.student.updateOne(
  {"student_id" : 1},
  {
    $set: {
      "firstname": "Adam",
      "lastname": "Silver"
    }
  }
)
```

```
db.student.updateOne(
  {"student_id" : 1},
  {
    $inc: {"age": 5}
  }
)
```

```
db.student.updateOne(
  {"student_id" : 1},
  {
    $set: {
      "firstname": "John",
      "lastname": "Gold"
    },
    $inc: {
      "age": -2
    }
  }
)
```

```
db.student.updateOne(
  {"student_id" : 2},
  {
    $set: {
      "age": 19
    }
  }
)
```

```
db.student.updateOne(
  {"student_id" : 2},
  {
    $unset: {
      "age": ""
    }
  }
)
```

- usuń wybrane dokumenty
 - po wykonaniu każdego z przykładów sprawdź wynik za pomocą `db.student.find()`
- np

```
db.student.deleteOne({"student_id": 1})
db.student.deleteMany({"student_id": 1})
db.student.deleteOne({"_id": ObjectId("67e669647013d10c86e71c87")});
```

- `replace`

```
db.student.find({"student_id": 1});
```

```
db.student.replaceOne(
  {
    "student_id": 1
  },
  {
    "student_id": 1,
    "firstname": "Jan",
    "lastname": "Kowalski",
    "hobby": "Ski"
  }
);
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "hobby": "Ski",
    "lastname": "Kowalski",
    "firstlame": "Jan",
    "student_id": 1
  }
]
```

- tablice
 - `grades` - tablica ocen studenta

```
db.student.updateOne(
  {
    "student_id": 1
  },
  {
    $set : {
      "grades" : [3, 4, 5, 6, 8]
    }
  }
);
```

```
db.student.find({"student_id": 1});
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
    "grades": [3, 4, 5, 6, 8],
    "hobby": "Ski",
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

- średnia ocena

```
db.student.aggregate([
  {
    $match : {
      "student_id": 1
    }
  },
  {
    $project: {
      "firstname": 1,
      "lastname": 1,
      "student_id": 1,
      "grades": 1,
      "averagegrade": { $avg: "$grades" }
    }
  }
])
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "averagegrade": 5.2,
    "firstname": "Jan",
    "grades": [3, 4, 5, 6, 8],
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

- dodanie elementu do tablicy

```
db.student.updateOne(
  { student_id: 1 },
  { $push: { grades: 10 } }
);
```

```
db.student.find({"student_id": 1});
```

- wynik


```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
    "grades": [3, 4, 5, 6, 8, 10],
    "hobby": "Ski",
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

- modyfikacja elementu w tablicy
 - indeksy zaczynają się od 0

```
db.student.updateOne(
  { student_id: 1 },
  { $set: { "grades.2": 12 } }
)
```

```
db.student.find({"student_id": 1});
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
    "grades": [3, 4, 12, 6, 8, 10],
    "hobby": "Ski",
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

- `$map` - wykonanie operacji dla każdego elementu tablicy
 - <https://www.mongodb.com/docs/manual/reference/operator/aggregation/map/>
 - w przykładzie poniżej
 - dzielimy każdą ocenę przez 20

```
db.student.updateOne(
  { student_id: 1 },
```

```
[
  {
    $set: {
      grades: {
        $map: {
          input: "$grades",
          as: "g",
          in: { $divide: ["$$g", 20] }
        }
      }
    }
  }
]
```

```
db.student.find({"student_id": 1});
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
    "grades": [0.15, 0.2, 0.6, 0.3, 0.4, 0.5],
    "hobby": "Ski",
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

Zadanie 1

- Wykonaj kilka własnych eksperymentów z operacjami CRUD
- przydatne linki
 - <https://www.mongodb.com/docs/manual/crud/>
 - <https://www.mongodb.com/docs/manual/reference/method/db.collection.find/>
 - <https://www.mongodb.com/docs/manual/reference/method/db.collection.insertOne/>
 - <https://www.mongodb.com/docs/manual/reference/method/db.collection.updateOne/>
 - <https://www.mongodb.com/docs/manual/reference/method/db.collection.deleteOne/>

Zadanie 1 - rozwiązanie

Wyniki:

przykłady, kod, zrzuty ekranów, komentarz ...

```
//dodanie do bazy
db.students.insertOne({
  name: "Anna",
  age: 22,
  major: "Computer Science",
  grades: [5, 4, 5]
})
db.students.insertMany([
  { name: "Kamil", age: 24, major: "Math", grades: [3, 4, 4] },
  { name: "Ola", age: 21, major: "Physics", grades: [5, 5, 5] }
])
```

```
//wynik
{
  _id: ObjectId('69ca85d2586a94aa2a433460'),
  name: 'Anna',
  age: 22,
  major: 'Computer Science',
  grades: [
    5,
    4,
    5
  ]
}

{
  _id: ObjectId('69ca85ea586a94aa2a433461'),
  name: 'Kamil',
  age: 24,
  major: 'Math',
  grades: [
    3,
    4,
    4
  ]
}

{
  _id: ObjectId('69ca85ea586a94aa2a433462'),
  name: 'Ola',
  age: 21,
  major: 'Physics',
  grades: [
    5,
    5,
    5
  ]
}
```

Przykład 2

- wybierz bazę `employees`

```
use employees;  
  
db;
```

- wstaw dokument do kolekcji `employees`

```
db.employees.insertOne(  
  {  
    "EmployeeID": 1,  
    "FirstName": "Nancy",  
    "LastName": "Davolio",  
    "Address" : {  
      "Street": "507 - 20th Ave. E. Apt. 2A",  
      "City": "Seattle",  
      "Country": "USA"  
    },  
    "Title": "Sales Representative",  
    "TitleOfCourtesy": "Ms.",  
    "BirthDate": ISODate("1948-12-08T00:00:00.000Z"),  
    "HireDate": ISODate("1992-05-01T00:00:00.000Z"),  
    "Phone": ["(206) 555-9857", "(206) 555-9858"],  
    "Salary": 1000  
  }  
)
```

- sprawdź zawartość kolekcji

```
db.employees.find();
```

- wstaw kilka kolejnych dokumentów za pomocą `insertMany`

```
db.employees.insertMany(  
  [  
    {  
      "EmployeeID": 2,  
      "FirstName": "Andrew",  
      "LastName": "Fuller",  
      "Address" : {
```

```

        "Street": "908 W. Capital Way",
        "City": "Tacoma",
        "Country": "USA",
    },
    "Title": "Vice President, Sales",
    "TitleOfCourtesy": "Dr.",
    "BirthDate": ISODate("1952-02-19T00:00:00.000Z"),
    "HireDate": ISODate("1992-08-14T00:00:00.000Z"),
    "Phone": ["(206) 555-9482"],
    "Salary": 10000
},
{
    "EmployeeID": 3,
    "FirstName": "Janet",
    "LastName": "Leverling",
    "Address" : {
        "Street": "722 Moss Bay Blvd.",
        "City": "Kirkland",
        "Country": "USA",
    },
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms.",
    "BirthDate": ISODate("1963-08-30T00:00:00.000Z"),
    "HireDate": ISODate("1992-04-01T00:00:00.000Z"),
    "Phone": ["(206) 555-3412"],
    "Salary": 1200
},
{
    "EmployeeID": 4,
    "FirstName": "Margaret",
    "LastName": "Peacock",
    "Address" : {
        "Street": "4110 Old Redmond Rd.",
        "City": "Redmond",
        "Country": "USA",
    },
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Mrs.",
    "BirthDate": ISODate("1937-09-19T00:00:00.000Z"),
    "HireDate": ISODate("1993-05-03T00:00:00.000Z"),
    "Phone": ["(206) 555-8122"],
    "Salary": 1100
},
{
    "EmployeeID": 5,
    "FirstName": "Steven",
    "LastName": "Buchanan",
    "Address" : {
        "Street": "14 Garrett Hill",
        "City": "London",
        "Country": "UK",
    },
    "Title": "Sales Manager",
    "TitleOfCourtesy": "Mr.",

```

```

    "BirthDate": ISODate("1955-03-04T00:00:00.000Z"),
    "HireDate": ISODate("1993-10-17T00:00:00.000Z"),
    "Phone": ["(71) 555-4848"],
    "Salary": 2000
  },
  {
    "EmployeeID": 6,
    "FirstName": "Michael",
    "LastName": "Suyama",
    "Address" : {
      "Street": "Coventry House Miner Rd.",
      "City": "London",
      "Country": "UK",
    },
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Mr.",
    "BirthDate": ISODate("1963-07-02T00:00:00.000Z"),
    "HireDate": ISODate("1993-10-17T00:00:00.000Z"),
    "Phone": ["(71) 555-7773"],
    "Salary": 1500
  },
  {
    "EmployeeID": 7,
    "FirstName": "Robert",
    "LastName": "King",
    "Address" : {
      "Street": "Edgeham Hollow Winchester Way",
      "City": "London",
      "Country": "UK",
    },
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Mr.",
    "BirthDate": ISODate("1960-05-29T00:00:00.000Z"),
    "HireDate": ISODate("1994-01-02T00:00:00.000Z"),
    "Phone": ["(71) 555-5598"],
    "Salary": 1000
  },
  {
    "EmployeeID": 8,
    "FirstName": "Laura",
    "LastName": "Callahan",
    "Address" : {
      "Street": "4726 - 11th Ave. N.E.",
      "City": "Seattle",
      "Country": "USA",
    },
    "Title": "Inside Sales Coordinator",
    "TitleOfCourtesy": "Ms.",
    "BirthDate": ISODate("1958-01-09T00:00:00.000Z"),
    "HireDate": ISODate("1994-03-05T00:00:00.000Z"),
    "Phone": ["(206) 555-1189"],
    "Salary": 3000
  },
  {

```

```

        "EmployeeID": 9,
        "FirstName": "Anne",
        "LastName": "Dodsworth",
        "Address" : {
            "Street": "7 Houndstooth Rd.",
            "City": "London",
            "Country": "UK",
        },
        "Title": "Sales Representative",
        "TitleOfCourtesy": "Ms.",
        "BirthDate": ISODate("1966-01-27T00:00:00.000Z"),
        "HireDate": ISODate("1994-11-15T00:00:00.000Z"),
        "Phone": ["(71) 555-4444"],
        "Salary": 1400
    }
]
)

```

- w **DataGrip** możesz posłużyć się poleceniami **SQL**
 - podzbiór poleceń **SQL** - proste polecenia **SELECT**
- wykonaj kilka takich poleceń
 - przeanalizuj wyniki
- np.

```

select * from employees;
select * from employees where Address.Country = "USA";
select * from employees where Address.Country = "USA" and TitleOfCourtesy = "Ms.";

select * from employees where Salary > 1200;

```

- napisz odpowiadające im polecenia **find**
- np

```

db.employees.find({})
db.employees.find({"Address.Country": "USA"})
db.employees.find({$and: [{"Address.Country": "USA"}, {"TitleOfCourtesy" :
"Ms."}]})

db.employees.find(
  {
    $and: [
      {"Address.Country": "USA"},
      {"TitleOfCourtesy" : "Ms."}
    ]
  }
)

```

```
)  
  
db.employees.find({ Salary: { $gt: 1200 } })
```

- operacja projekcji (wybór atrybutów w zbiorze wynikowym)

```
db.employees.find({})
```

```
db.employees.find(
  {},
  {"FirstName": 1, "LastName": 1}
)
```

- wynik

```
[
{
  "_id": {"$oid": "67e879dbb1c69e3d7e42c9c3"},
  "FirstName": "Nancy",
  "LastName": "Davolio"
},
{
  "_id": {"$oid": "67e879e2b1c69e3d7e42c9c5"},
  "FirstName": "Andrew",
  "LastName": "Fuller"
},
{
  "_id": {"$oid": "67e879e2b1c69e3d7e42c9c6"},
  "FirstName": "Janet",
  "LastName": "Leverling"
},
{
  "_id": {"$oid": "67e879e2b1c69e3d7e42c9c7"},
  "FirstName": "Margaret",
  "LastName": "Peacock"
},
{
  "_id": {"$oid": "67e879e2b1c69e3d7e42c9c8"},
  "FirstName": "Steven",
  "LastName": "Buchanan"
},
{
  "_id": {"$oid": "67e879e2b1c69e3d7e42c9c9"},
  "FirstName": "Michael",
  "LastName": "Suyama"
},
{
```



```

    "_id": {"$oid": "67e879e2b1c69e3d7e42c9ca"},
    "FirstName": "Robert",
    "LastName": "King"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9cb"},
    "FirstName": "Laura",
    "LastName": "Callahan"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9cc"},
    "FirstName": "Anne",
    "LastName": "Dodsworth"
  }
]

```

- warunek i projekcja

```

db.employees.find(
  {
    "Address.Country": "USA"
  },
  {
    "FirstName": 1,
    "LastName": 1
  }
)

```

- wynik

```

[
  {
    "FirstName": "Nancy",
    "LastName": "Davolio"
  },
  {
    "FirstName": "Andrew",
    "LastName": "Fuller"
  },
  {
    "FirstName": "Janet",
    "LastName": "Leverling"
  },
  {
    "FirstName": "Margaret",
    "LastName": "Peacock"
  },
  {
    "FirstName": "Laura",

```

```

    "LastName": "Callahan"
  }
]

```

- inne przykłady
 - 1 - atrybut pojawi się w zbiorze wynikowym
 - 0 - atrybut nie pojawi się
 - mogą wystąpić albo 1 albo 0
 - nie dotyczy atrybutu `_id`

```

db.employees.find({}, {"_id": 0, "FirstName": 1, "LastName": 1 })

db.employees.find({}, {"_id": 1, "FirstName": 0, "LastName": 0})

// to jest błąd
db.employees.find({}, {"_id": 1, "FirstName": 1, "LastName": 0})

```

- `count`

```

select count(*) from employees;
select count(*) from employees where Address.Country = "USA";
select count(*) from employees where Address.Country = "USA" and TitleOfCourtesy = "Ms.";

select count(*) from employees where Salary > 1200;

```

```

db.employees.find({}).count();
db.employees.find({"Address.Country": "USA"}).count();

db.employees.find({$and: [{"Address.Country": "USA"}, {"TitleOfCourtesy" : "Ms."}]}).count();

db.employees.find(
  {
    $and: [
      {"Address.Country": "USA"},
      {"TitleOfCourtesy" : "Ms."}
    ]
  }
).count();

```

```

db.employees.find({ Salary: { $gt: 1200 } }).count();

```

- wynik

```
[
  {
    "result": 5
  }
]
```

- \$or \$and

```
select * from employees where Salary > 1100 and Salary < 1500
```

```
db.employees.find({
  Salary: { $gt: 1100, $lt: 1500 }
})

db.employees.find(
  {
    $and: [
      { Salary: { $gt: 1100 } },
      { Salary: { $lt: 1500 } }
    ]
  }
);
```

- wynik

```
[
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c6"},
    "Address": {
      "Street": "722 Moss Bay Blvd.",
      "City": "Kirkland",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1963-08-30T00:00:00.000Z"},
    "EmployeeID": 3,
    "FirstName": "Janet",
    "HireDate": {"$date": "1992-04-01T00:00:00.000Z"},
    "LastName": "Leverling",
    "Phone": ["(206) 555-3412"],
    "Salary": 1200,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms."
  },
  {
```

```

    "_id": {"$oid": "67e879e2b1c69e3d7e42c9cc"},
    "Address": {
      "Street": "7 Houndstooth Rd.",
      "City": "London",
      "Country": "UK"
    },
    "BirthDate": {"$date": "1966-01-27T00:00:00.000Z"},
    "EmployeeID": 9,
    "FirstName": "Anne",
    "HireDate": {"$date": "1994-11-15T00:00:00.000Z"},
    "LastName": "Dodsworth",
    "Phone": ["(71) 555-4444"],
    "Salary": 1400,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms."
  }
]

```

```
select * from employees where Salary <= 1100 or Salary >= 1500
```

```

db.employees.find(
  {
    $or: [
      { Salary: { $lte: 1100 } },
      { Salary: { $gte: 1500 } }
    ]
  }
);

```

- wynik

```

[
  {
    "_id": {"$oid": "67e879dbb1c69e3d7e42c9c3"},
    "Address": {
      "Street": "507 - 20th Ave. E.\r\nApt. 2A",
      "City": "Seattle",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1948-12-08T00:00:00.000Z"},
    "EmployeeID": 1,
    "FirstName": "Nancy",
    "HireDate": {"$date": "1992-05-01T00:00:00.000Z"},
    "LastName": "Davolio",
    "Phone": ["(206) 555-9857", "(206) 555-9858"],
    "Salary": 1000,
    "Title": "Sales Representative",
  }
]

```

```

    "TitleOfCourtesy": "Ms."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c5"},
    "Address": {
      "Street": "908 W. Capital Way",
      "City": "Tacoma",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1952-02-19T00:00:00.000Z"},
    "EmployeeID": 2,
    "FirstName": "Andrew",
    "HireDate": {"$date": "1992-08-14T00:00:00.000Z"},
    "LastName": "Fuller",
    "Phone": ["(206) 555-9482"],
    "Salary": 10000,
    "Title": "Vice President, Sales",
    "TitleOfCourtesy": "Dr."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c7"},
    "Address": {
      "Street": "4110 Old Redmond Rd.",
      "City": "Redmond",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1937-09-19T00:00:00.000Z"},
    "EmployeeID": 4,
    "FirstName": "Margaret",
    "HireDate": {"$date": "1993-05-03T00:00:00.000Z"},
    "LastName": "Peacock",
    "Phone": ["(206) 555-8122"],
    "Salary": 1100,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Mrs."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c8"},
    "Address": {
      "Street": "14 Garrett Hill",
      "City": "London",
      "Country": "UK"
    },
    "BirthDate": {"$date": "1955-03-04T00:00:00.000Z"},
    "EmployeeID": 5,
    "FirstName": "Steven",
    "HireDate": {"$date": "1993-10-17T00:00:00.000Z"},
    "LastName": "Buchanan",
    "Phone": ["(71) 555-4848"],
    "Salary": 2000,
    "Title": "Sales Manager",
    "TitleOfCourtesy": "Mr."
  },
  {

```

```

    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c9"},
    "Address": {
      "Street": "Coventry House Miner Rd.",
      "City": "London",
      "Country": "UK"
    },
    "BirthDate": {"$date": "1963-07-02T00:00:00.000Z"},
    "EmployeeID": 6,
    "FirstName": "Michael",
    "HireDate": {"$date": "1993-10-17T00:00:00.000Z"},
    "LastName": "Suyama",
    "Phone": ["(71) 555-7773"],
    "Salary": 1500,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Mr."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9ca"},
    "Address": {
      "Street": "Edgeham Hollow Winchester Way",
      "City": "London",
      "Country": "UK"
    },
    "BirthDate": {"$date": "1960-05-29T00:00:00.000Z"},
    "EmployeeID": 7,
    "FirstName": "Robert",
    "HireDate": {"$date": "1994-01-02T00:00:00.000Z"},
    "LastName": "King",
    "Phone": ["(71) 555-5598"],
    "Salary": 1000,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Mr."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9cb"},
    "Address": {
      "Street": "4726 - 11th Ave. N.E.",
      "City": "Seattle",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1958-01-09T00:00:00.000Z"},
    "EmployeeID": 8,
    "FirstName": "Laura",
    "HireDate": {"$date": "1994-03-05T00:00:00.000Z"},
    "LastName": "Callahan",
    "Phone": ["(206) 555-1189"],
    "Salary": 3000,
    "Title": "Inside Sales Coordinator",
    "TitleOfCourtesy": "Ms."
  }
]

```

```
select * from employees where year(HireDate) = 1992
```

```
db.employees.find(
  {
    $expr: {
      $eq: [ { $year: "$HireDate" }, 1992 ]
    }
  }
);

db.employees.find(
  {
    HireDate: {
      $gte: ISODate("1992-01-01T00:00:00Z"),
      $lt: ISODate("1993-01-01T00:00:00Z")
    }
  }
);
```

- wynik

```
[
  {
    "_id": {"$oid": "67e879dbb1c69e3d7e42c9c3"},
    "Address": {
      "Street": "507 - 20th Ave. E.\r\nApt. 2A",
      "City": "Seattle",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1948-12-08T00:00:00.000Z"},
    "EmployeeID": 1,
    "FirstName": "Nancy",
    "HireDate": {"$date": "1992-05-01T00:00:00.000Z"},
    "LastName": "Davolio",
    "Phone": ["(206) 555-9857", "(206) 555-9858"],
    "Salary": 1000,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c5"},
    "Address": {
      "Street": "908 W. Capital Way",
      "City": "Tacoma",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1952-02-19T00:00:00.000Z"},
```

```
"EmployeeID": 2,
"FirstName": "Andrew",
"HireDate": {"$date": "1992-08-14T00:00:00.000Z"},
"LastName": "Fuller",
"Phone": ["(206) 555-9482"],
"Salary": 10000,
"Title": "Vice President, Sales",
"TitleOfCourtesy": "Dr."
},
{
  "_id": {"$oid": "67e879e2b1c69e3d7e42c9c6"},
  "Address": {
    "Street": "722 Moss Bay Blvd.",
    "City": "Kirkland",
    "Country": "USA"
  },
  "BirthDate": {"$date": "1963-08-30T00:00:00.000Z"},
  "EmployeeID": 3,
  "FirstName": "Janet",
  "HireDate": {"$date": "1992-04-01T00:00:00.000Z"},
  "LastName": "Leverling",
  "Phone": ["(206) 555-3412"],
  "Salary": 1200,
  "Title": "Sales Representative",
  "TitleOfCourtesy": "Ms."
}
]
```

Zadanie 2

- Wykonaj kilka własnych eksperymentów

Zadanie 2 - rozwiązanie

Wyniki:

przykłady, kod, zrzuty ekranów, komentarz ...

-- ...

Operacje agregacji

- aggregate
- aggregation pipeline
- <https://www.mongodb.com/docs/manual/core/aggregation-pipeline/>


```
db.<collection>.aggregate(  
  
  [  
  
    {stage 1},  
  
    {stage 2},  
  
    ... *  
  
    {stage N}  
  
  ]  
  
)
```

- <https://www.mongodb.com/docs/manual/reference/operator/aggregation-pipeline/#std-label-aggregation-pipeline-operator-reference>
- Stages
 - `$match`
 - `$project`
 - `$group`
 - `$unwind`
 - `$lookup`
 - `$out`
 - ...
- Aggregation vs SQL
 - <https://www.mongodb.com/docs/manual/reference/sql-aggregation-comparison/>

Przykład 3

- baza `employees` z poprzedniego przykładu

```
use employees;  
  
db;  
  
db.employees.find();
```

- Operacje agregacji
 - `aggregate`

```
db.employees.aggregate([
  {
    $match: {
      "Address.Country": "USA"
    }
  },
  {
    $project: {
      "_id": 0,
      "LastName": 1,
      "FirstName": 1}
  },
  {
    $sort: {LastName: -1}
  }
])
```

- wynik

```
[
  {
    "FirstName": "Nancy",
    "LastName": "Davolio"
  },
  {
    "FirstName": "Andrew",
    "LastName": "Fuller"
  },
  {
    "FirstName": "Janet",
    "LastName": "Leverling"
  },
  {
    "FirstName": "Margaret",
    "LastName": "Peacock"
  },
  {
    "FirstName": "Laura",
    "LastName": "Callahan"
  }
]
```

- dodatkowe pole `FullName`

```
db.employees.aggregate([
  {
    $match: {
      "Address.Country": "USA"
    }
  }
```

```

    },
    {
      $project: {
        "_id": 0,
        "First": "$LastName",
        "Last": "$LastName",
        "FullName": {
          $concat: ["$FirstName", " ", "$LastName"]
        }
      }
    },
    {
      $sort: {lastname: -1}
    }
  ]
})

```

- wynik

```

[
  {
    "First": "Davolio",
    "FullName": "Nancy Davolio",
    "Last": "Davolio"
  },
  {
    "First": "Fuller",
    "FullName": "Andrew Fuller",
    "Last": "Fuller"
  },
  {
    "First": "Leverling",
    "FullName": "Janet Leverling",
    "Last": "Leverling"
  },
  {
    "First": "Peacock",
    "FullName": "Margaret Peacock",
    "Last": "Peacock"
  },
  {
    "First": "Callahan",
    "FullName": "Laura Callahan",
    "Last": "Callahan"
  }
]

```

- grupowanie
 - \$group

```
select TitleOfCourtesy, count(*)
from employees
where Address.Country = "USA"
group by TitleOfCourtesy
```

```
db.employees.aggregate([
  {
    $match: {
      "Address.Country": "USA"
    }
  },
  {
    $group: {
      "_id": "$TitleOfCourtesy",
      "cnt": {"$sum": 1}
    }
  }
])
```

- wynik

```
[
  {
    "_id": "Dr.",
    "cnt": 1
  },
  {
    "_id": "Mrs.",
    "cnt": 1
  },
  {
    "_id": "Ms.",
    "cnt": 3
  }
]
```

- dodatkowa projekcja, zmiana nazwy

```
db.employees.aggregate([
  {
    $match: {
      "Address.Country": "USA"
    }
  },
  {
    $group: {
```

```

        "_id": "$TitleOfCourtesy",
        "cnt": {"$sum": 1}
    }
},
{
    $project: {
        "_id": 0,
        "TitleOfCourtesy": "$_id",
        "cnt": 1}
    },
])

```

- grupowanie wg kilku atrybutów

```

select Address.Country, TitleOfCourtesy, count(*) cnt
from employees
group by Address.Country, TitleOfCourtesy
order by cnt desc;

```

```

db.employees.aggregate([
    {
        $group: {
            _id: {
                country: "$Address.Country",
                title: "$TitleOfCourtesy"
            },
            cnt: { $sum: 1 }
        }
    },
    {
        $project: {
            _id: 0,
            country: "$_id.country",
            title: "$_id.title",
            cnt: 1
        }
    },
    {
        $sort: {cnt: -1}
    }
])

```

- wynik

```

[
  {

```

```

    "Country": "UK",
    "TitleOfCourtesy": "Mr.",
    "cnt": 3
  },
  {
    "Country": "USA",
    "TitleOfCourtesy": "Ms.",
    "cnt": 3
  },
  {
    "Country": "USA",
    "TitleOfCourtesy": "Dr.",
    "cnt": 1
  },
  {
    "Country": "UK",
    "TitleOfCourtesy": "Ms.",
    "cnt": 1
  },
  {
    "Country": "USA",
    "TitleOfCourtesy": "Mrs.",
    "cnt": 1
  }
]

```

- zapis wyniku
 - \$out
 - zapis zbioru wynikowego to kolekcji o wskazanej nazwie

```

db.employees.aggregate([
  {
    $group: {
      _id: {
        Country: "$Address.Country",
        TitleOfCourtesy: "$TitleOfCourtesy"
      },
      cnt: { $sum: 1 }
    }
  },
  {
    $project: {
      _id: 0,
      Country: "$_id.Country",
      TitleOfCourtesy: "$_id.TitleOfCourtesy",
      cnt: 1
    }
  },
  {
    $out: "em_by_country_title"
  }
]

```

```
}  
])
```

- lista kolekcji

```
show collections;
```

- wynik

```
[  
  {  
    "badge": "",  
    "name": "em_by_country_title"  
  },  
  {  
    "badge": "",  
    "name": "employees"  
  }  
]
```

- zawartość kolekcji `em_by_cuntry_title`

```
db.em_by_country_title.find();
```

- wynik

```
[  
  {  
    "_id": {"$oid": "67ec20f77b1ef941c9512dcb"},  
    "Country": "USA",  
    "TitleOfCourtesy": "Mrs.",  
    "cnt": 1  
  },  
  {  
    "_id": {"$oid": "67ec20f77b1ef941c9512dcc"},  
    "Country": "USA",  
    "TitleOfCourtesy": "Ms.",  
    "cnt": 3  
  },  
  {  
    "_id": {"$oid": "67ec20f77b1ef941c9512dcd"},  
    "Country": "USA",  
    "TitleOfCourtesy": "Dr.",  
    "cnt": 1  
  },  
]
```

```
{
  "_id": {"$oid": "67ec20f77b1ef941c9512dce"},
  "Country": "UK",
  "TitleOfCourtesy": "Mr.",
  "cnt": 3
},
{
  "_id": {"$oid": "67ec20f77b1ef941c9512dcf"},
  "Country": "UK",
  "TitleOfCourtesy": "Ms.",
  "cnt": 1
}
]
```

Zadanie 3

- Wykonaj kilka własnych eksperymentów

Zadanie 3 - rozwiązanie

Wyniki:

przykłady, kod, zrzuty ekranów, komentarz ...

-- ...

Przykład 4

Odtwórz z backupu bazę `north0`

- plik z z backup'em bazy danych dostępny jest w moodle
- najprostsza wersja
 - `mongorestore <folder>`

```
mongorestore dump
```

- to polecenie odtworzy wszystkie bazy danych znajdujące się we wskazanym folderze (w tym przypadku `dump`)
 - najłatwiej wgrać tam folder zawierający pliki z backupem i wykonać proste polecenie `mongorestore`
- dokumentacja:
 - <https://www.mongodb.com/docs/database-tools/mongorestore/>

Wybierz bazę north0

```
use north0
```

Baza **north0** jest kopią relacyjnej bazy danych **Northwind**

- poszczególne kolekcje odpowiadają tabelom w oryginalnej bazie **Northwind**

zapoznaj się ze strukturą dokumentów w bazie **North0**

```
db.customers.find()  
db.orders.find();  
db.orderdetails.find();
```

Przykład 5

produkty, kategorie, dostawcy

```
db.categories.find().limit(2);
```

- wynik

```
[  
  {  
    "_id": {"$oid": "67eb06a27b1ef941c950fb49"},  
    "CategoryID": 1,  
    "CategoryName": "Beverages",  
    "Description": "Soft drinks, coffees, teas, beers, and ales"  
  },  
  {  
    "_id": {"$oid": "67eb06a27b1ef941c950fb4a"},  
    "CategoryID": 2,  
    "CategoryName": "Condiments",  
    "Description": "Sweet and savory sauces, relishes, spreads, and seasonings"  
  }  
]
```

```
db.products.find().limit(2);
```

- wynik

```
[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "SupplierID": 1,
    "UnitPrice": 18,
    "UnitsInStock": 39,
    "UnitsOnOrder": 0
  },
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 2,
    "ProductName": "Chang",
    "QuantityPerUnit": "24 - 12 oz bottles",
    "ReorderLevel": 25,
    "SupplierID": 1,
    "UnitPrice": 19,
    "UnitsInStock": 17,
    "UnitsOnOrder": 40
  }
]
```

```
db.suppliers.find().limit(2);
```

- wynik

```
[
  {
    "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
    "Address": "49 Gilbert St.",
    "City": "London",
    "CompanyName": "Exotic Liquids",
    "ContactName": "Charlotte Cooper",
    "ContactTitle": "Purchasing Manager",
    "Country": "UK",
    "Fax": null,
    "HomePage": null,
    "Phone": "(171) 555-2222",
    "PostalCode": "EC1 4SD",
  }
]
```

```

    "Region": null,
    "SupplierID": 1
  },
  {
    "_id": {"$oid": "63a05d3cbb3b972d6f4e09f8"},
    "Address": "P.O. Box 78934",
    "City": "New Orleans",
    "CompanyName": "New Orleans Cajun Delights",
    "ContactName": "Shelley Burke",
    "ContactTitle": "Order Administrator",
    "Country": "USA",
    "Fax": null,
    "HomePage": "#CAJUN.HTM#",
    "Phone": "(100) 555-4822",
    "PostalCode": "70117",
    "Region": "LA",
    "SupplierID": 2
  }
]

```

Uwaga:

- etap `$match` z pustym warunkiem zwraca wszystkie dokumenty (nie jest potrzebny, można go pominąć)
 - w przykładzie został dodany żeby łatwiej można było wykonać własne eksperymenty ograniczające zbiór dokumentów z kolekcji `products`

```

db.products.aggregate([
  {
    $match : { }
  },
  //suppliers
  {
    $lookup : {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier"
    }
  },
  {
    $limit : 2
  }
])

```

- pole `Supplier` jest tablicą
 - zawiera 1 obiekt
- wynik

```
[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "Supplier": [
      {
        "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
        "SupplierID": 1,
        "CompanyName": "Exotic Liquids",
        "ContactName": "Charlotte Cooper",
        "ContactTitle": "Purchasing Manager",
        "Address": "49 Gilbert St.",
        "City": "London",
        "Region": null,
        "PostalCode": "EC1 4SD",
        "Country": "UK",
        "Phone": "(171) 555-2222",
        "Fax": null,
        "HomePage": null
      }
    ],
    "SupplierID": 1,
    "UnitPrice": 18,
    "UnitsInStock": 39,
    "UnitsOnOrder": 0
  },
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 2,
    "ProductName": "Chang",
    "QuantityPerUnit": "24 - 12 oz bottles",
    "ReorderLevel": 25,
    "Supplier": [
      {
        "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
        "SupplierID": 1,
        "CompanyName": "Exotic Liquids",
        "ContactName": "Charlotte Cooper",
        "ContactTitle": "Purchasing Manager",
        "Address": "49 Gilbert St.",
        "City": "London",
        "Region": null,
        "PostalCode": "EC1 4SD",
        "Country": "UK",
        "Phone": "(171) 555-2222",
        "Fax": null,

```

```

    "HomePage": null
  }
],
"SupplierID": 1,
"UnitPrice": 19,
"UnitsInStock": 17,
"UnitsOnOrder": 40
}
]
```

- \$unwind
 - pole **Supplier** jest obiektem/dokumentem

```

db.products.aggregate([
  {
    $match : { }
  },

  //suppliers
  {
    $lookup : {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier"
    }
  },
  {
    $unwind : "$Supplier"
  },
  {
    $limit : 2
  }
])
```

- wynik

```

[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "Supplier": {
      "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
      "SupplierID": 1,

```

```

    "CompanyName": "Exotic Liquids",
    "ContactName": "Charlotte Cooper",
    "ContactTitle": "Purchasing Manager",
    "Address": "49 Gilbert St.",
    "City": "London",
    "Region": null,
    "PostalCode": "EC1 4SD",
    "Country": "UK",
    "Phone": "(171) 555-2222",
    "Fax": null,
    "HomePage": null
  },
  "SupplierID": 1,
  "UnitPrice": 18,
  "UnitsInStock": 39,
  "UnitsOnOrder": 0
},
{
  "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
  "CategoryID": 1,
  "Discontinued": false,
  "ProductID": 2,
  "ProductName": "Chang",
  "QuantityPerUnit": "24 - 12 oz bottles",
  "ReorderLevel": 25,
  "Supplier": {
    "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
    "SupplierID": 1,
    "CompanyName": "Exotic Liquids",
    "ContactName": "Charlotte Cooper",
    "ContactTitle": "Purchasing Manager",
    "Address": "49 Gilbert St.",
    "City": "London",
    "Region": null,
    "PostalCode": "EC1 4SD",
    "Country": "UK",
    "Phone": "(171) 555-2222",
    "Fax": null,
    "HomePage": null
  },
  "SupplierID": 1,
  "UnitPrice": 19,
  "UnitsInStock": 17,
  "UnitsOnOrder": 40
}
]

```

- projekcja
 - tylko wybrane pola z **Supplier**

```

db.products.aggregate([
  {
    $match : { }
  },
  //suppliers
  {
    $lookup : {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier_tmp"
    }
  },
  {
    $unwind : "$Supplier_tmp"
  },
  {
    $addFields: {
      Supplier: {
        SupplierID: "$SupplierID",
        CompanyName: "$Supplier_tmp.CompanyName",
        Country: "$Supplier_tmp.Country"
      }
    }
  },
  {
    $project: {
      SupplierID: 0,
      Supplier_tmp: 0
    }
  },
  {
    $limit : 2
  }
])

```

- wynik

```

[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "Supplier": {
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",

```

```

    "Country": "UK"
  },
  "UnitPrice": 18,
  "UnitsInStock": 39,
  "UnitsOnOrder": 0
},
{
  "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
  "CategoryID": 1,
  "Discontinued": false,
  "ProductID": 2,
  "ProductName": "Chang",
  "QuantityPerUnit": "24 - 12 oz bottles",
  "ReorderLevel": 25,
  "Supplier": {
    "SupplierID": 1,
    "CompanyName": "Exotic Liquids",
    "Country": "UK"
  },
  "UnitPrice": 19,
  "UnitsInStock": 17,
  "UnitsOnOrder": 40
}
]

```

- dodaj do wyniku inf. o kategoriach

```

db.products.aggregate([
  {
    $match : { }
  },
  //suppliers
  {
    $lookup : {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier_tmp"
    }
  },
  //categories
  {
    $lookup : {
      from: "categories",
      localField: "CategoryID",
      foreignField: "CategoryID",
      as: "Category_tmp"
    }
  },
  {

```



```

        $unwind : "$Supplier_tmp"
    },
    {
        $unwind : "$Category_tmp"
    },
    {
        $addFields: {
            Supplier: {
                SupplierID: "$SupplierID",
                CompanyName: "$Supplier_tmp.CompanyName",
                Country: "$Supplier_tmp.Country"
            },
            Category: {
                CategoryID: "$CategoryID",
                CategorynameName: "$Category_tmp.CategoryName",
            }
        }
    },
    {
        $project: {
            SupplierID: 0,
            Supplier_tmp: 0,
            CategoryID: 0,
            Category_tmp: 0
        }
    },
    {
        $limit : 2
    }
])

```

- wynik

```

[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "Category": {
      "CategoryID": 1,
      "CategorynameName": "Beverages"
    },
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "Supplier": {
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",
      "Country": "UK"
    },
    "UnitPrice": 18,
    "UnitsInStock": 39,
  }
]

```

```

    "UnitsOnOrder": 0
  },
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
    "Category": {
      "CategoryID": 1,
      "CategorynameName": "Beverages"
    },
    "Discontinued": false,
    "ProductID": 2,
    "ProductName": "Chang",
    "QuantityPerUnit": "24 - 12 oz bottles",
    "ReorderLevel": 25,
    "Supplier": {
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",
      "Country": "UK"
    },
    "UnitPrice": 19,
    "UnitsInStock": 17,
    "UnitsOnOrder": 40
  }
]

```

Przykład 6

- dostawcy i produkty

```

db.suppliers.aggregate([
  {
    $match : { }
  },
  //products
  {
    $lookup : {
      from: "products",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Products"
    }
  },
  {
    $addFields: {
      Products: {
        $map: {
          input: "$Products",
          as: "product",
          in: {

```

```

        ProductID: "$$product.ProductID",
        ProductName: "$$product.ProductName",
        UnitPrice: "$$product.UnitPrice",
        Discontinued: "$$product.Discontinued",
    }
}
}
},
{
    $limit : 2
}
])

```

- wynik

```

[
  {
    "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
    "Address": "49 Gilbert St.",
    "City": "London",
    "CompanyName": "Exotic Liquids",
    "ContactName": "Charlotte Cooper",
    "ContactTitle": "Purchasing Manager",
    "Country": "UK",
    "Fax": null,
    "HomePage": null,
    "Phone": "(171) 555-2222",
    "PostalCode": "EC1 4SD",
    "Products": [
      {
        "ProductID": 1,
        "ProductName": "Chai",
        "UnitPrice": 18,
        "Discontinued": false
      },
      {
        "ProductID": 2,
        "ProductName": "Chang",
        "UnitPrice": 19,
        "Discontinued": false
      },
      {
        "ProductID": 3,
        "ProductName": "Aniseed Syrup",
        "UnitPrice": 10,
        "Discontinued": false
      }
    ],
    "Region": null,
    "SupplierID": 1
  },
]

```

```
{
  "_id": {"$oid": "63a05d3cbb3b972d6f4e09f8"},
  "Address": "P.O. Box 78934",
  "City": "New Orleans",
  "CompanyName": "New Orleans Cajun Delights",
  "ContactName": "Shelley Burke",
  "ContactTitle": "Order Administrator",
  "Country": "USA",
  "Fax": null,
  "HomePage": "#CAJUN.HTM#",
  "Phone": "(100) 555-4822",
  "PostalCode": "70117",
  "Products": [
    {
      "ProductID": 4,
      "ProductName": "Chef Anton's Cajun Seasoning",
      "UnitPrice": 22,
      "Discontinued": false
    },
    {
      "ProductID": 5,
      "ProductName": "Chef Anton's Gumbo Mix",
      "UnitPrice": 21.35,
      "Discontinued": true
    },
    {
      "ProductID": 65,
      "ProductName": "Louisiana Fiery Hot Pepper Sauce",
      "UnitPrice": 21.05,
      "Discontinued": false
    },
    {
      "ProductID": 66,
      "ProductName": "Louisiana Hot Spiced Okra",
      "UnitPrice": 17,
      "Discontinued": false
    }
  ],
  "Region": "LA",
  "SupplierID": 2
}
```

Przykład 7

- zamówienia

```
db.orders.aggregate(
  {
```

```
    $match : { }
  },

  //customers
  {
    $lookup : {
      from: "customers",
      localField: "CustomerID",
      foreignField: "CustomerID",
      as: "Customer"
    }
  },
  {
    $unwind : "$Customer"
  },
  {
    $project : {
      "Customer._id": 0,
      "Customer.ContactName": 0,
      "Customer.ContactTitle": 0,
      "Customer.City": 0,
      //      "Customer.Country": 0,
      "Customer.Address": 0,
      "Customer.PostalCode": 0,
      "Customer.Region": 0,
      "Customer.Phone": 0,
      "Customer.Fax": 0
    }
  },

  //employees
  {
    $lookup : {
      from: "employees",
      localField: "EmployeeID",
      foreignField: "EmployeeID",
      as: "Employee"
    }
  },
  {
    $unwind : "$Employee"
  },
  {
    $project : {
      "Employee._id": 0,
      "Employee.Title": 0,
      "Employee.TitleOfCourtesy": 0,
      "Employee.BirthDate": 0,
      "Employee.HireDate": 0,
      "Employee.Address": 0,
      "Employee.PostalCode": 0,
      "Employee.City": 0,
      "Employee.Region": 0,
```

```

        "Employee.Country": 0,
        "Employee.HomePhone": 0,
        "Employee.Extension": 0,
        "Employee.Photo": 0,
        "Employee.Notes": 0,
        "Employee.ReportsTo": 0,
        "Employee.PhotoPath": 0
    }
},
//shippers
{
    $lookup : {
        from: "shippers",
        localField: "ShipVia",
        foreignField: "ShipperID",
        as: "Shipper"
    }
},
{
    $unwind : "$Shipper"
},
{
    $project : {
        "Shipper._id": 0,
        "Shipper.Phone": 0
    }
},
//orderdetails
{
    $lookup : {
        from: "orderdetails_tmp",
        localField: "OrderID",
        foreignField: "OrderID",
        as: "Orderdetails"
    }
},
{
    $project : {
        "Orderdetails._id": 0,
        "Orderdetails.OrderID": 0
    }
},
{
    $project : {
        "CustomerID": 0,
        "EmployeeID": 0,
        "ShipVia": 0
    }
},
{
    $out : "orders_tmp"
}

```

```
)
```

```
db.orders_tmp.find().limit(2);
```

- wynik

```
[
  {
    "_id": {"$oid": "63a060b9bb3b972d6f4e1fc6"},
    "Customer": {
      "CustomerID": "VINET",
      "CompanyName": "Vins et alcools Chevalier",
      "Country": "France"
    },
    "Employee": {
      "EmployeeID": 5,
      "LastName": "Buchanan",
      "FirstName": "Steven"
    },
    "Freight": 32.38,
    "OrderDate": {"$date": "1996-07-04T00:00:00.000Z"},
    "OrderID": 10248,
    "Orderdetails": [
      {
        "UnitPrice": 14,
        "Quantity": 12,
        "Discount": 0,
        "product": {
          "ProductID": 11,
          "ProductName": "Queso Cabrales",
          "QuantityPerUnit": "1 kg pkg."
        },
        "TotalValue": 168
      },
      {
        "UnitPrice": 9.8,
        "Quantity": 10,
        "Discount": 0,
        "product": {
          "ProductID": 42,
          "ProductName": "Singaporean Hokkien Fried Mee",
          "QuantityPerUnit": "32 - 1 kg pkgs."
        },
        "TotalValue": 98
      },
      {
        "UnitPrice": 34.8,
        "Quantity": 5,
```

```

    "Discount": 0,
    "product": {
      "ProductID": 72,
      "ProductName": "Mozzarella di Giovanni",
      "QuantityPerUnit": "24 - 200 g pkgs."
    },
    "TotalValue": 174
  }
],
"RequiredDate": {"$date": "1996-08-01T00:00:00.000Z"},
"ShipAddress": "59 rue de l'Abbaye",
"ShipCity": "Reims",
"ShipCountry": "France",
"ShipName": "Vins et alcools Chevalier",
"ShipPostalCode": "51100",
"ShipRegion": null,
"ShippedDate": {"$date": "1996-07-16T00:00:00.000Z"},
"Shipper": {
  "ShipperID": 3,
  "CompanyName": "Federal Shipping"
}
},
{
  "_id": {"$oid": "63a060b9bb3b972d6f4e1fc7"},
  "Customer": {
    "CustomerID": "TOMSP",
    "CompanyName": "Toms Spezialitäten",
    "Country": "Germany"
  },
  "Employee": {
    "EmployeeID": 6,
    "LastName": "Suyama",
    "FirstName": "Michael"
  },
  "Freight": 11.61,
  "OrderDate": {"$date": "1996-07-05T00:00:00.000Z"},
  "OrderID": 10249,
  "Orderdetails": [
    {
      "UnitPrice": 18.6,
      "Quantity": 9,
      "Discount": 0,
      "product": {
        "ProductID": 14,
        "ProductName": "Tofu",
        "QuantityPerUnit": "40 - 100 g pkgs."
      },
      "TotalValue": 167.4
    },
    {
      "UnitPrice": 42.4,
      "Quantity": 40,
      "Discount": 0,
      "product": {

```



```

        "ProductID": 51,
        "ProductName": "Manjimup Dried Apples",
        "QuantityPerUnit": "50 - 300 g pkgs."
      },
      "TotalValue": 1696
    }
  ],
  "RequiredDate": {"$date": "1996-08-16T00:00:00.000Z"},
  "ShipAddress": "Luisenstr. 48",
  "ShipCity": "Münster",
  "ShipCountry": "Germany",
  "ShipName": "Toms Spezialitäten",
  "ShipPostalCode": "44087",
  "ShipRegion": null,
  "ShippedDate": {"$date": "1996-07-10T00:00:00.000Z"},
  "Shipper": {
    "ShipperID": 1,
    "CompanyName": "Speedy Express"
  }
}
]

```

Zadanie 4

- zmień (popraw) strukturę dokumentu w kolekcji `orders_tmp`
- stwórz obiekt/dokument grupujący inf. o adresie wysyłki
 - np.

```

Shippment : {
  "ShipAddress": "Luisenstr. 48",
  "ShipCity": "Münster",
  "ShipCountry": "Germany",
  "ShipName": "Toms Spezialitäten",
  "ShipPostalCode": "44087",
  "ShipRegion": null,
}

```

- stwórz obiekt/dokument grupujący inf. o datach (zamówienia, wysyłki ...)
 - np.

```

Dates : {
  "OrderDate": {"$date": "1996-07-04T00:00:00.000Z"},
  "RequiredDate": {"$date": "1996-08-01T00:00:00.000Z"},
  "ShippedDate": {"$date": "1996-07-16T00:00:00.000Z"},
}

```

- Dodaj inf. o pełnej wartości całego zamówienia

TotalOrderValue : ...

Zadanie 4 - rozwiązanie

Wyniki:

przykłady, kod, zrzuty ekranów, komentarz ...

```
db.orders_tmp.updateMany(
  {},
  [
    {
      $set: {
        Shipment: {
          ShipAddress: "$ShipAddress",
          ShipCity: "$ShipCity",
          ShipCountry: "$ShipCountry",
          ShipName: "$ShipName",
          ShipPostalCode: "$ShipPostalCode",
          ShipRegion: "$ShipRegion"
        }
      }
    },
    {
      $unset: [
        "ShipAddress",
        "ShipCity",
        "ShipCountry",
        "ShipName",
        "ShipPostalCode",
        "ShipRegion"
      ]
    }
  ]
);
```

```
db.orders_tmp.updateMany(
  {},
  [
    {
      $set: {
        Dates: {
          OrderDate: "$OrderDate",
          RequiredDate: "$RequiredDate",
          ShippedDate: "$ShippedDate"
        }
      }
    }
  ]
);
```

```

    }
  },
  {
    $unset: [
      "OrderDate",
      "RequiredDate",
      "ShippedDate"
    ]
  }
]
);
db.orders_tmp.updateMany(
  {},
  [
    {
      $set: {
        TotalOrderValue: {
          $sum: {
            $map: {
              input: "$OrderDetails",
              as: "item",
              in: {
                $multiply: [
                  "$$item.UnitPrice",
                  "$$item.Quantity",
                  { $subtract: [1, "$$item.Discount"] }
                ]
              }
            }
          }
        }
      }
    }
  ]
);

```



```
> db.orders_tmp.findOne();

< {
  _id: ObjectId('69dc375d9dea0b1969aeca08'),
  OrderID: 10248,
  CustomerID: 'VINET',
  EmployeeID: 5,
  OrderDetails: [
    {
      ProductID: 11,
      UnitPrice: 14,
      Quantity: 12,
      Discount: 0
    },
    {
      ProductID: 42,
      UnitPrice: 9.8,
      Quantity: 10,
      Discount: 0
    },
    {
      ProductID: 72,
      UnitPrice: 34.8,
      Quantity: 5,
```

```
    Discount: 0.1
  }
],
```

```
Shippment: {
  ShipAddress: "59 rue de l'Abbaye",
  ShipCity: 'Reims',
  ShipCountry: 'France',
  ShipName: 'Vins et alcools Chevalier',
  ShipPostalCode: '51100',
  ShipRegion: null
},
Dates: {
  OrderDate: 1996-07-04T00:00:00.000Z,
  RequiredDate: 1996-08-01T00:00:00.000Z,
  ShippedDate: 1996-07-16T00:00:00.000Z
},
TotalOrderValue: 422.6
}
```

shop>